FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –1
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. ABC

Sol. $-k\upsilon = \frac{d\upsilon}{dt}$, $\int \frac{d\upsilon}{\upsilon} = -k\ dt \int_t^t dt$

$\ln \frac{\upsilon}{\upsilon_0} = -kt$ $\qquad \upsilon = \upsilon_0 e^{-kt}$

$\upsilon_0 e^{-kt} = \frac{dx}{dt}$

$\int_0^x dx = \upsilon_0 \int_0^t e^{-kt} dt$

$x = \frac{\upsilon_0}{k}\left[e^{-kt}\right]_0^t = -\frac{\upsilon_0}{k}\left(e^{-kt} - 1\right)$, $x = \frac{\upsilon_0}{k}\left(1 - e^{-kt}\right)$

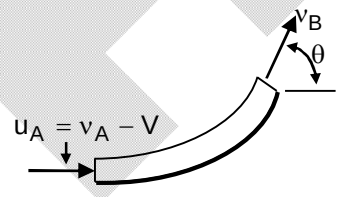
$-k\upsilon = \upsilon\frac{d\upsilon}{dx}$, $d\upsilon = -k\ dx$

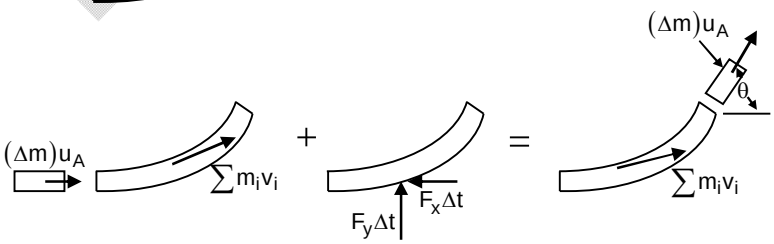
$\int_{\upsilon_0}^{\upsilon} d\upsilon = -k\int_0^x dx$

$\upsilon - \upsilon_0 = -kx$, $\upsilon = \upsilon_0 - kx$

2. ABD

Sol.





$\xrightarrow{+}$ x components: $(\Delta m)u - F_x\Delta t = (\Delta m)u\cos\theta$

$+\uparrow$ y components: $+F_y\Delta t = (\Delta m)u\sin\alpha$

Substituting $\Delta m = A\rho(\upsilon_A - V)\Delta t$ and $u = \upsilon_A - V$, we obtain

$F_x = A\rho(\upsilon_A - V)^2(1-\cos\theta) \leftarrow$ and $F_y = A\rho(\upsilon A - V)^2\sin\theta\uparrow$

Power $= F_xV = A\rho(\upsilon_A - V)^2(1-\cos\theta)V$

$\frac{d(\text{power})}{dV} = A\rho\left(\upsilon_A^2 - 4\upsilon_A V + 3V^2\right)(1-\cos\theta) = 0$

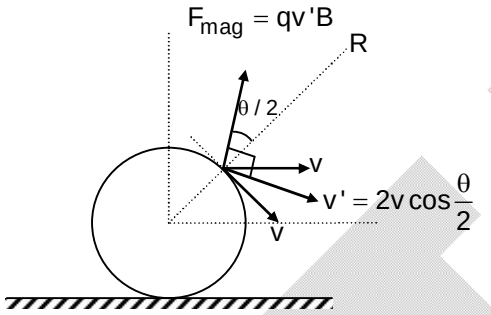
$V = \upsilon_A \quad V = \frac{V_A}{3}$ For maximum power $V = \frac{V_A}{3}$

3. ACD

Sol. $f = qv'B\ \sin\frac{\theta}{2} = q\left\{2v\cos\left(\frac{\theta}{2}\right)\right\}B\sin\frac{\theta}{2} = Bqv\sin\theta$

Since, speed at topmost point is maximum

$\therefore\ N = mg - 2Bqv$



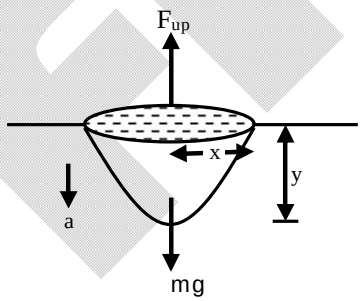
$\Rightarrow$ To loose the contact, $N = 0$

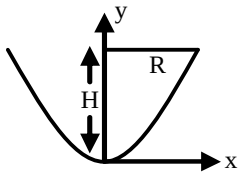
$\Rightarrow\ v = \frac{mg}{2qB}$

4. AD

Sol. Let $Y = ax^2$ (A parabola)

$Y = \frac{H}{R^2}x^2$ (from given data)





$v = \frac{dy}{dt}$

$$mg-\rho_L\cdot\frac{1}{2}\pi x^2 y\cdot g=m\cdot\frac{dv}{dt}$$

$$\frac{dv}{dt}=g-\frac{\pi x^2 yg\rho_L}{2\rho_s\frac{\pi R^2 H}{2}}g\Rightarrow g-\frac{\rho_L}{\rho_s}\frac{x^2}{R^2}\times\frac{yg}{H}$$

$$v\cdot\frac{dv}{dy}=g-\frac{\rho_L}{\rho_s}\frac{gy^2}{H^2}$$

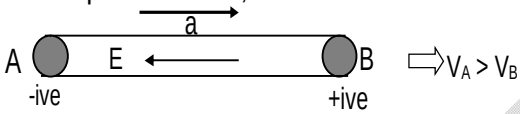
$$\int_0^0 vdv=\int_0^H\left(g-\frac{\rho_L}{\rho_s}\frac{gy^2}{H^2}\right)dy$$

$$o=gH-\frac{g\rho_L}{3\rho_s}H$$

$$\frac{\rho_L}{\rho_s}=3$$

5. ABCD

Sol. Due to pseudo force, electrons move from B to A.



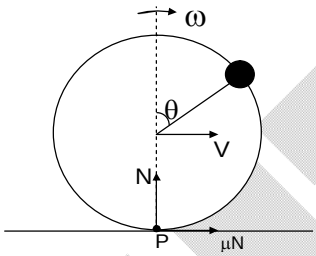
$\Rightarrow V_A > V_B$

In figure (b) electron moves from A to B

Region I Region II

6. BC

Sol.



V_C is Horizontal only.

$\overrightarrow{V_P}=\overrightarrow{V_{PC}}+\overrightarrow{V_C}$

$V_P=0$

Mechanical energy conservation,

$$mgR(1-\cos\theta)=\frac{1}{2}mV_{bead}^2 \qquad V_{Bead}=2v\cos\theta/2$$

$$gR\times 2\sin^2\theta/2=2v^2\cos^2\theta/2$$

$$V=\sqrt{gR}\tan\theta/2$$

$\overrightarrow{\tau_{em}}=I_{cm}\propto=0$ As $I_{cm}=0$

$$NR\sin\theta=\mu N(R+R\cos\theta)$$

$$\tan\theta/2=\mu$$

$$\theta=2\tan^{-1}(\mu)$$

7. C

Sol. For He^+ absorption occur from level $(2\rightarrow 4)$

For Li^{2+} absorption occur from level $(3\rightarrow 6)$

8. D

Sol. (I) If 1st case $E = E_1 = E_0 \sin\omega t$

If 2nd case $E = E_2 = E_0 \sin(\omega t + \phi)$

If 3rd case $E = E_1 + E_2 = E_0 '\sin(\omega t + \delta)$

All three cases effective value is same $\Rightarrow E_0 ' = E_0$

$$\Rightarrow \qquad \phi = \frac{2\pi}{3}$$

(II) Case : (2) When S_3, S_4 closed $Z_1 = \sqrt{2}R$

Case : (2) When all opened $Z_2 = \sqrt{R^2 + (X_C - X_L)^2}$

$$\Rightarrow \qquad X_C - X_L = R \Rightarrow X_C = 2R \Rightarrow \frac{X_C}{R} = 2$$

$$\Rightarrow \qquad \frac{4\pi}{5}\frac{X_C}{R} = \frac{8\pi}{5}$$

(III) When S_1, S_2 closed $Z_1 = \sqrt{5}R$

When S_3, S_4 closed $Z_2 = \sqrt{2}R$

$$\frac{I_1}{I_2} = \sqrt{\frac{2}{5}} \Rightarrow \pi\left(\frac{I_1}{I_2}\right) = \frac{2\pi}{5}$$

(IV) When all opened $Z_1 = \sqrt{2}R$

When S_2, S_3 Closed $Z_2 = R$

$$\frac{I_3}{I_4} = \frac{1}{\sqrt{2}} \Rightarrow \pi\left(\frac{I_3}{I_4}\right)^2 = \frac{\pi}{2}$$

9. B

Sol. (I) The fundamental frequency in the string.

$$f_0 = \frac{\sqrt{T/\mu}}{2\ell} = \sqrt{\frac{102.4}{1\times10^{-3}}} \times \frac{1}{2\times0.5}\text{Hz} = 320 \text{ Hz.}$$

Other possible resonance frequencies are f_A and $f_0 = 320$ Hz, 640 Hz, 960 Hz.

(II) The fundamental frequency in the string.

$$f_0 = \frac{\sqrt{T/\mu}}{4\ell} = \frac{320}{4\times0.5} = 160 \text{ Hz.}$$

Other possible resonance frequencies are

$f_B = 160$ Hz, 480 Hz, 800 Hz.

(III) The fundamental frequency in both ends open organ pipe is

$$f_0 = \frac{v}{2\ell} = \frac{320}{2\times0.5} = 320 \text{ Hz.}$$

Other possible resonance frequencies are

$f_c = 320$ Hz, 640 Hz, 960 Hz

(IV) The fundamental frequency in one end open organ pipe is

$$f_0 = \frac{v}{4\ell} = \frac{320}{4\times0.5} = 160 \text{ Hz.}$$

Other possible resonance frequencies are

$f_D = 160$ Hz, 480 Hz, 800 Hz.

10. B

Sol. $E_x = \frac{-1}{a}$ $E_y = \frac{-2V}{b}$ $E_z = \frac{-3V}{c}$

$\Delta V = -\vec{E}.\vec{d}$

Section – B

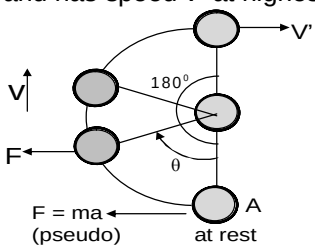
11. 25.00

Sol. Acceleration of train = $a \longrightarrow$

Motion of bob w.r.t. train:

Initially $\vec{V}_{bob/train} = O$

From $A \longrightarrow B$, pseudo force F = ma $\longleftarrow$ acts on bob. Let bob turns through 180° and has speed v' at highest position



W/E theorem $A \to B$:

$FR = \frac{1}{2}mv^2 + mgh$

$\frac{mv^2}{R} = 2(ma - mg)$ .....(i)

$B \longrightarrow C$: Now train has stopped, a = 0,

No pseudo force acts on bob.

Conservation of Energy

$mgR + \frac{1}{2}mv'^2 = \frac{1}{2}mv^2$

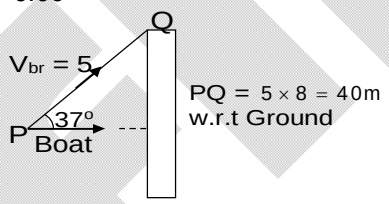
$\frac{mv'^2}{R} = \frac{mv^2}{R} - 2mg$

T + mg = 2 (ma – mg) – 2mg [From eq. (i)]

On solving $a \geq 25 m/s^2$

12. 6.00

Sol.

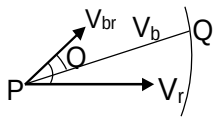


$\overrightarrow{V_b} = \overrightarrow{V_{br}} + \overrightarrow{V_r}$

Velocity must be along PQ

$v_{br} \sin\theta = v_r \sin 37$

$\sin\theta = 7/25$



$$t = \frac{40}{v_{br}\cos\theta + v_r \cos 37^\circ} = 6 \text{ sec.}$$

13. 25.00

Sol. $I_1 + I_2 = 13I_0$

$$\left(\sqrt{I_2} - \sqrt{I_2}\right)^2 = I_0$$

$$I_1 + I_2 - 2\sqrt{I_1}\sqrt{I_2} = I_0$$

$$2\sqrt{I_1}\sqrt{I_2} = 12I_0$$

$$I_{max} = 13I_0 + 12I_0 = 25I_0 = (5)^2 I_0$$

14. 0.50

Sol. $\phi = \left[\vec{E}.\vec{n}\right]A = \left[\frac{8\times 2 + (6\times 3) + (10\times 6)}{\sqrt{2^2+3^2+6^2}}\right]\frac{7}{47}$

15. 41.23

Sol. $\tan\alpha = \frac{v_y}{v_x}$

16. 1.80

Sol. $V_I - \frac{q_1}{7} - V_G = 0$

$$V_{IG} = \frac{1.80}{11}$$

17. 2.75

Sol. Equivalent resistance across capacitor (after shorting the battery) $= \frac{11R}{4}$

Voltage across capacitor in steady state $= \frac{3E}{4}$

$$q = \frac{CE}{4}\left(1 - e^{\frac{-4t}{11RC}}\right)$$

18. 5.94

Sol. Least count $= \frac{P}{N} = \frac{1\text{mm}}{100} = 0.01\,\text{mm}$

Diameter of the wire is

$d = 4\text{mm} + 56\times \text{L.C} - 6\times \text{L.C}$

$d = 4\text{mm} + 50\times \text{L.C}$

$d = 4\text{mm} + 50\times 0.01\text{mm}$

$d = 4.50\,\text{mm}$

$d = 0.450$ cm

The curved surface area of the wire is

$S = \pi dl$

$$S = \frac{22}{7}\times 0.450 \times 4.2 = 5.94\,\text{cm}^2$$

Chemistry

PART – II

Section – A

19. ABC

Sol. $(x-1)\left(x^2-8x+12\right)=0$

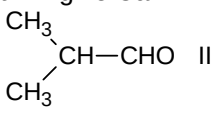
$x=1,\ 2,\ 6$

$x=\dfrac{2r}{a_0}\Big|r=\dfrac{a_0}{2},\ a_0,\ 3a_0$

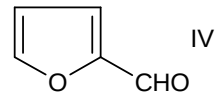
Minimum position	maximum position
$0.5\ a_0$	$3a_0$

20. BCD

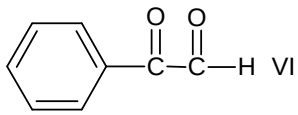
Sol. It will give Cannizzaro because it fails to give self aldol



II



IV



VI

These will give Cannizzaro reaction.

21. AD

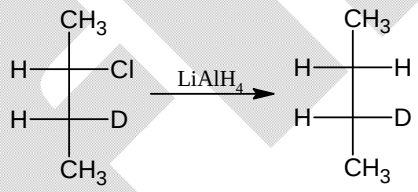
Sol. $C_2=\sigma1s^2\ \overset{*}{\sigma}1s^2\sigma2s^2\ \overset{*}{\sigma}2s^2\ \underbrace{\pi2p_x^2\pi2p_y^2}_{\text{forms }\pi\text{ bond}}$

22. BCD

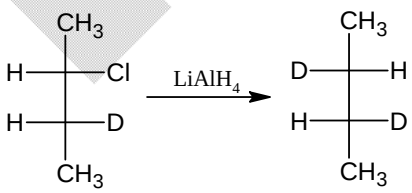
Sol. Fact

23. AB

Sol.



A — Optically active



A — Optically active

24. ABCD

Sol. (A) NH_3 has higher boiling point than AsH_3 and PH_3 on account H-bonding. SbH_3 has highest boiling point on account of highest molecular weight.

(B) As the size of element increases, the strength of M-H bond decreases and thus stability decreases.

(C) Basicity decreases down the group. (Drago rule).

(D) (Drago rule). HEH angle $(^\circ): NH_3 = 107.8; PH_3 = 93.6, AsH_3 = 91.8; SbH_3 = 91.3$.

25. A

Sol. Electrolysis of NaCl

(I) Fused NaCl

(II) Conc. $NaCl_{(aq)}$

(III) Dil. $NaCl_{(aq)}$

(IV) NaCl (mercury electrode)

(I) Fused NaCl:

At cathode:

$$Na^+ + e^- \longrightarrow Na_{(s)} \quad ;E^\circ = -2.71V$$

At anode:

$$2Cl^- \longrightarrow Cl_{2(g)} + 2e^- \quad ;E^\circ = -1.36V$$

Voltage $= -2.71 - 1.36 = -4.07V$

(II) Conc. $NaCl_{(aq)}$

At cathode:

$$2H^+ - 2e^- \longrightarrow H_{2(g)} \quad ;E^\circ = 0.00V$$

$$2H_2O + 2e^- \longrightarrow H_{2(g)} + 20H^- \quad ;E^\circ = -0.828V$$

$$Na^+ + e^- \longrightarrow Na \quad ;E^\circ = -2.71V$$

At anode:

$$40H^- \longrightarrow O_2 + 2H_2O + 4e^- \quad ;E^\circ = -0.401V$$

$$2H_2O \longrightarrow O_2 + 4H^+ + 4e^- \quad ;E^\circ = -1.229V$$

$$T,K\ 2Cl^- \longrightarrow Cl_2 + 2e^- \quad ;E^\circ = -1.36V$$

(due to over voltage and high conc. of Cl^-)

At cathode: $H_{2(g)}$

At anode: $Cl_{2(g)}$, [O_2 also but percentage may be low]

To increase the kinetics of reactions a higher voltage (as from required) is provided, due to this Cl^- start to oxidized at the anode, So

Cl_2 – higher voltage

O_2 – lower voltage

(III) Dil. $NaCl_{(aq)}$

At cathode:

$$2H_2O + 2e^- \longrightarrow H_{2(g)} + 20H^- \quad ;E^\circ = -0.828V$$

At anode:

$$2H_2O \longrightarrow O_{2(g)} + 4H^+ + 4e^- \quad ;E^\circ = -1.229V$$

At cathode: $H_{2(g)}$

At anode: $O_{2(g)}$

(IV) $NaCl_{(aq)}$ (Mercury electrode):
As Na makes an amalgam with Hg so the kinetics favoured the occurrence of Na rather than $H_{2(g)}$.

26. D
Sol. Fact

27. A
Sol.

(I) $CH_2{=}CH{-}CH{=}O \xrightarrow[alc.]{NaBH_4} CH_2{=}CH{-}CH_2OH$
Unsaturated alcohol

(II) $C_6H_5CH{=}O + H_2N{-}Ph \xrightarrow{pH(4\text{-}5)} C_6H_5CH{=}N{-}Ph$
Schiff's base

(III) $C_6H_5(H_3C)C{=}O + HN(CH_3)(CH_2CH_3) \xrightarrow{pH(4\text{-}5)} C_6H_5(H_2C{=})C{-}N(CH_3)(CH_2CH_3)$
Enamine

(IV) $R{-}CH{=}O + 2RCH_2 \xrightarrow{H^+} R{-}HC(OCH_2R)(OCH_2R)$
Acetal

28. C
Sol.

(I) $(H_3C)_3C{-}Cl + EtO^{\ominus}Na^{+} \longrightarrow (H_3C)(CH_3)C{=}CH_2 + EtOH$
Alcohol
Q

(II) $CH_3CH_2Cl + Me_3CO^{\ominus}Na^{\oplus} \longrightarrow CH_3CH_2{-}O{-}C(Me)_3 + CH_2{=}CH_2$
P

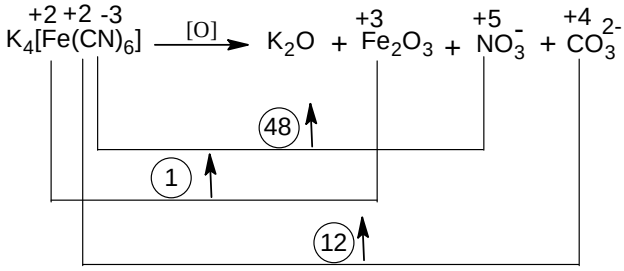
(III) $C_6H_5{-}Cl + MeO^{\ominus}Na^{+} \longrightarrow$ NO rxn

(IV) $C_6H_5{-}O^{\ominus}Na^{+} + Me{-}Cl \longrightarrow C_6H_5{-}O{-}CH_3$
Anisole (R)

Section – B

29. 61.00

Sol.



Total $\uparrow$ in oxidation number $= 48 + 12 + 1 = 61$

So 'n' factor $= 61$

30. 25.00

Sol. $\underset{1-0.6}{HIn} \rightleftharpoons H^+ + \underset{0.6}{In^-}$

$= 0.4$

$$pH = pK_{In} + \log\frac{\left[In^-\right]}{[HIn]}$$

$$2.176 = pK_{In} + \log\frac{0.6}{0.4}$$

$$pK_{In} = 2 \Rightarrow K_{In} = 10^{-2}$$

$\underset{1-0.6}{HIn} \rightleftharpoons H^+ + \underset{0.6}{In^-}$

$= 0.4$

now $\underset{C(1-\alpha)}{HIn} \rightleftharpoons \underset{3.33\times10^{-3}}{H^+} + \underset{C\alpha}{In^-}$

$$\frac{3.33\times10^{-3}\times C\alpha}{C(1-\alpha)} = 10^{-2}$$

$$\frac{\alpha}{1-\alpha} = 3 \quad \Rightarrow \alpha = 3 - 3\alpha$$

$$\Rightarrow \alpha = 0.75$$

So $[HIn] = 1 - 0.75 = 0.25$

$\%[HIn] = 25\%$

31. 15.38

Sol. Total work done in the cyclic process

= Area under PV

$= P_0V_0$

Now,

$$A \xrightarrow[\text{Heat taken}]{} B \xrightarrow[\text{Heat taken}]{} C \xrightarrow[\text{Heat rejected}]{} D \xrightarrow[\text{Heat rejected}]{} A$$

$= \underset{\text{at constant volume}}{1\times\frac{3R}{2}(2T-T)} \Big| \underset{\text{at constant pressure}}{=1\times\frac{3R}{2}(4T-2T)}$

$= \frac{3}{2}RT$ $\frac{5}{2}\times 2RT$

$= 1.5RT$ $5RT$

$= 1.5P_0V_0$ $5P_0V_0$ | At point A $P_0V_0 = 1RT$

Total Heat taken

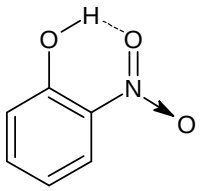
$= 6.5\ P_0V_0$

$\eta = \frac{\text{work done}}{\text{Heat taken}} = \frac{P_0V_0}{6.5P_0V_0} = 0.1538$ i.e. 15.38%.

32. 600.00

Sol. o-nitrophenol p-nitrophenol

Note:



Due to intramolecular H Bonding $V\cdot P$ of O nitrophenol will be higher. If one have $V\cdot P = 400$ & the $V\cdot P$ of mixture is 520 mm of Hg, i.e. the $V\cdot P$ of other will be higher than 400, so if P is the $V\cdot P$ then it will of O nitrophenol

O nitrophenol P nitrophenol

P 400

$520 = P\times 0.6 + 400\times 0.4$

$P = \frac{520-160}{0.6} = \frac{360}{0.6} = 600$ mm of Hg.

33. 122.40

Sol. $25.7 = 13.6\ z^2\left[\frac{1}{2^2}-\frac{1}{n^2}\right]$ .....(i)

$8.7 = 13.6\ z^2\left[\frac{1}{3^2}-\frac{1}{n^2}\right]$ .....(ii)

Now (i) ÷ (ii)

$\frac{25.7}{8.7} = \frac{\left(n^2-4\right)9n^2}{4n^2\left(n^2-9\right)}$

$\frac{25.7}{8.7} = \frac{9n^2-36}{4n^2-36}$

$2.95\times 4n^2 - 2.95\times 36 = 9n^2 - 36$

$n^2\left(11.816-9\right) = 36\times 1.95$

$n^2 = 25 \quad \Rightarrow \quad n = 5$

Now,

$25.7 = 13.6z^2\left[\frac{1}{4}-\frac{1}{25}\right]$

$z = 3$

Ionisation energy $= 13.6\times 3^2 = 122.4$ eV

34. 25.00

Sol. $CuSO_2 \cdot 5H_2O \rightleftharpoons CuSO_4 \cdot 3H_2O_{(s)} + \underset{p}{2H_2O_{(g)}}$

$kp = p^2$

$10^{-4} = p^2$

$p = 10^{-2}$ atm

$= 10^{-2} \times 760 = 7.6$ mm of Hg

Relative humidity $= \frac{7.6}{30.4} \times 100 = 25\%$

35. 915.00

Sol. In BCC nearest neighbor distance

$$= \frac{\sqrt{3}a}{2}$$

$$\frac{\sqrt{3}a}{2} = 4.52$$

$$a = \left(\frac{2 \times 4.52}{\sqrt{3}}\right) A^\circ$$

$$d = \frac{ZM}{NV} = \frac{2 \times 39}{6 \times 10^{23} \times \left(\frac{2 \times 4.52}{\sqrt{3}}\right)^3 \times 10^{-24}} = \frac{2 \times 39}{6 \times 14.2} = 0.915 \text{ g/cc}$$

$$= 915 \text{ kg/m}^3$$

36. 7482.60

Sol. Given: $P_1 = 16$ bar, $T_1 = 300$ K, $P_2 = 1$ bar, $T_2 = ?$

To calculate T_2 use:

$$TP^{\left(\frac{1-\gamma}{\gamma}\right)} = \text{cons t}$$

$$T_1P_1^{\left(\frac{1-\gamma}{\gamma}\right)} = T_2P_2^{\left(\frac{1-\gamma}{\gamma}\right)}$$

$$300 \times 16^{\left(\frac{1-\gamma}{\gamma}\right)} = T_2 \times 1^{\left(\frac{1-\gamma}{\gamma}\right)}$$

$$T_2 = 300 \times 16^{\left(\frac{3}{4}-1\right)}$$

$$T_2 = \frac{300}{2} = 150 \text{ K}$$

$$W_{rev.adia} = nC_V(T_2 - T_1)$$

$$= 2 \times 3R(150 - 300)$$

$$= -7482.6 \text{ J}$$

Mathematics

PART – III

Section – A

37. BC

Sol. $x^4(x-5)=1$

$\Rightarrow x-5=\frac{1}{x^4}$

$\therefore \sum\frac{1}{\alpha_r^4}=\sum\alpha_r-25=5-25=-20$

Similarly, $\left(\frac{1}{\alpha_r^4}+5\right)=\alpha_r$

38. A

Sol. $\because \begin{vmatrix}1 & a & -1\\ 2 & -1 & 3\\ a & -1 & 3\end{vmatrix}=0$

$\Rightarrow a=\frac{1}{3}$ or 2

But for a = 2, two vectors are parallel.

39. D

Sol. $\because I=\int_\alpha^\beta f(x)dx$

$\Rightarrow I=\int_\alpha^\beta f(\alpha+\beta-x)dx=\int_\alpha^\beta f(3-x)dx$

So, $7I=\int_\alpha^\beta\left(3x^2-8\right)dx$

$=\beta^3-\alpha^3-8(\beta-\alpha)$

$=(\beta-\alpha)\{\beta^2+\alpha^2+\beta\alpha-8\}$

$=\sqrt{9-4a}\{9-\alpha\beta-8\}=\sqrt{9-4a}(1-a)\le 0$

$\Rightarrow 1\le a\le\frac{9}{4}$

40. AB

Sol. If $\lambda x^2+(\lambda-1)x+\frac{1}{24}=0$ has equal and real roots,

$\Rightarrow(\lambda-1)^2=4.\lambda.\frac{1}{24}$

$\Rightarrow 6\lambda^2-13\lambda+6=0$

$\therefore \lambda=\frac{13\pm\sqrt{169-144}}{2\times 6}=\frac{2}{3}$ or $\frac{3}{2}$

41. BD

Sol. $\lim\limits_{n\to\infty}\left(\frac{1}{n^n}\right)^n.\left(n^n\right)^n\left[\left(1+\frac{1}{n}\right)\left(1+\frac{1}{2n}\right)\left(1+\frac{1}{2^2n}\right).....\right]^n$

$= e^1.e^{\frac{1}{2}}.e^{\frac{1}{2^2}}.....\infty$

$= e^{1+\frac{1}{2}+\frac{1}{2^2}+.....} = e^{\frac{1}{1-\frac{1}{2}}} = e^2 \simeq 7.3441$

42. AB

Sol. $|z_1| = |z_2| = 1$ & $|z_1 + z_2| = 1$

$\therefore |z_1 + z_2| = |z_1||1+e^{i\theta}| = 1$

$|z_1 - z_2| = |z_1||1-e^{i\theta}| = 1$

$\because z_2 = z_1\omega$

$\therefore z_1 + z_2 = z_1(1+\omega) = \frac{-b}{a}$

& $z_1z_2 = z_1^2\omega = \frac{c}{a}$

$\Rightarrow \left(\frac{b^2}{a^2}\right) = \frac{c}{a} \Rightarrow b^2 = ac$

$1+\cos^2\theta + 2\cos\theta + \sin^2\theta = 1$

$2 + 2\cos\theta = 1$

$\cos\theta = \frac{-1}{2} \Rightarrow \theta = \frac{2\pi}{3}$

43. C

Sol. AEEI (4 vowels) and DRJLNG (6 consonants)

$\therefore N = \frac{\underline{|4}}{\underline{|2}} \times \underline{|6} = 12 \times 720$

$= 2^6 \times 3^3 \times 5^1$

44. A

Sol. Either increasing or decreasing part lie in [0, 2] $\rightarrow$ [0, 1]

Then $y(0) = c = 1$ or 0

and $y = x(ax+b)+c \Rightarrow y(2) = 2(2a+b)+c = 1$ or 0

$\therefore |2a+b| = \frac{1}{2}$

If $\frac{\lambda}{a}$ is root, $a\times\frac{\lambda^2}{a^2} + b\times\frac{\lambda}{a} + c = 0 \Rightarrow \frac{\lambda^2}{a} + \frac{b\lambda}{a} + 1 = 0$ ..... (i)

$\because 4a + 2b + 1 = 0$

$1 + \frac{b}{2a} + \frac{1}{4a} = 0$ ..... (ii)

From (i) and (ii), $\lambda = \frac{1}{2}$

Sum of roots = $-\frac{b}{a}=\left(\frac{2a+\frac{1}{2}}{a}\right)=p^2+p+3\geq\frac{11}{4}$

$\because 4a+2b+1=0$ $\Rightarrow 2+\frac{1}{2a}\geq\frac{11}{4}$

$-b=\left(2a+\frac{1}{2}\right)$ For $a>0$, $\frac{1}{2a}\geq\frac{3}{4}\Rightarrow a\leq\frac{2}{3}$

For $a<0$, $a\geq\frac{2}{3}\Rightarrow|a|\leq\frac{2}{3}$

$|a|\leq 0.66$

45. D

Sol. $\because f(x)=\frac{2}{\pi}\tan\left(\frac{\pi x}{2}\right)+\frac{1}{2x}$ has point of minima at $x=\frac{1}{2}$

So, range $\in\left[\frac{\pi+2}{\pi},\infty\right)$

$\therefore e^{-k}\geq 1$

$\Rightarrow e^{k}\leq 1 \quad \Rightarrow k\leq 0$

So, for $k\in\left(-\frac{1}{e},0\right)$, we have two solutions.

In the same manner use graphs.

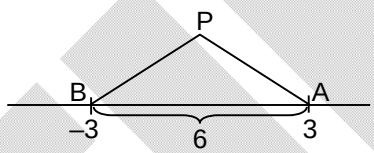
46. A

Sol. $\because z^2\bar{z}+\bar{z}^2z=(z+\bar{z})$

$\Rightarrow z\bar{z}(z+\bar{z})=(z+\bar{z})$

$\Rightarrow x^2+y^2=1$ or $x=0$

circle straight line



$|PA-PB|=6$

means, P lies on straight line

& $|PA-PB|<6$ shows it is hyperbola.

Section – B

47. 1.33

Sol. $\because \quad 1\leq|x|\leq\sqrt{3} \quad$ & $\quad \tan^{-1}\geq\frac{2\pi}{3}-\frac{\pi}{2}=\frac{\pi}{6}$

$\Rightarrow x\geq\frac{1}{\sqrt{3}} \quad \therefore \quad \frac{1}{3}\leq x\leq\sqrt{3}.$

$\therefore \quad \alpha+\frac{1}{\beta^2}=1+\frac{1}{3}=\frac{4}{3}.$

48. 0.25

Sol. $\because \quad y = \dfrac{x}{x^2+y} \qquad \Rightarrow y^2 + x^2y - x = 0$

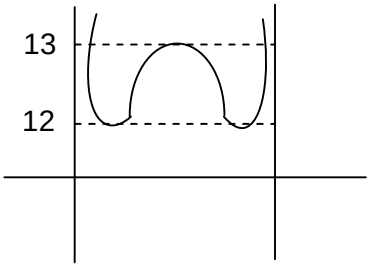
$$\therefore \quad \frac{dy}{dx} = \frac{-(2xy-1)}{(x^2+2y)} \Rightarrow \frac{dy}{dx} = \frac{-\left(2x.\dfrac{x}{x^2+y}-1\right)}{(x^2+y+y)} = -\frac{-(2x^2-x^2-y)/x^y y}{\left(\dfrac{x}{y}+y\right)}$$

$$= \frac{-y(x^2-y)}{(x^2+y)(x+y^2)} \Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{(y-x^2)}{(x^2+y)(x+y^2)}$$

$$= \frac{d}{dx}(\ln y) = \frac{(y-x^2)}{(x^2+y)(x+y^2)} \Rightarrow \ln y + c = \int \frac{(y-x^2)dx}{(x^2+y)(x+y^2)}$$

49. 4.50

Sol. Graph of $(9\sin x + 4\operatorname{cosec} x)$ is



$\because 12 < 2k + 4 \le 13$

$\Rightarrow 8 < 2k \le 9$

$4 < k \le \dfrac{9}{2}$

$\therefore k = \dfrac{9}{2}$

50. 1.78

Sol. (distance from focus)^2 = (Circumradius) $\times \dfrac{1}{4}$(L.R.).$\left(\dfrac{4}{3}\right)^2 = R \times \dfrac{1}{4} \times 4 \Rightarrow R = \dfrac{16}{9}$

51. 14.00

Sol. $\displaystyle\int_{-n}^{n} e^{\{|x|\}} \frac{dx}{1+e^{\sin x}}$

$= \displaystyle\int_0^n e^{\{|x|\}} dx.$

$= n\displaystyle\int_0^1 e^{x-[x]} = n(e-1)$

$\displaystyle\sum_1^{50} \frac{n}{(n+1)!} = 1 - \frac{1}{(3\times 17)!}$

So $\quad |a-b| = 14$

52. 1.00

Sol. $x^3 - x^2 - x + 1 = (x+1)(x-1)^2$, $f(x) = g(x)$ have 3 real and distinct roots

$\Rightarrow 0 < a^2 - 2 < 1$

$\Rightarrow [a^2] = 2$

Now $x^3 - x^2 - x + 1 = 2$
has solution in (1, 2)
$\Rightarrow$ the integral part of root is 1.

53. 4.00

Sol. $$\frac{3x^2+1}{\sqrt{\left(x^4+x^2\right)}} = \frac{\left(3x^2+1\right)}{\sqrt{x^2\left(x^2+1\right)}}$$

$$= \frac{2x^2+\left(x^2+1\right)}{x\sqrt{x^2+1}}$$

$$= \frac{2x}{\sqrt{x^2+1}} + \frac{\sqrt{x^2+1}}{x}$$

$\geq 2\sqrt{2}$

Equality for $2x^2 = x^2 + 1$

$x^2 = 1$

$x = 1$

$$\therefore \int_0^{\sqrt{2}} \frac{3x^2+1}{\sqrt{x^4+x^2}}\,dx \geq 2\sqrt{2}\left(\sqrt{2}-0\right) = 4$$

54. 147.00

Sol. Required probability = $\frac{176}{10\times10\times10} = \frac{22}{125}$

So, $p + q = 147$